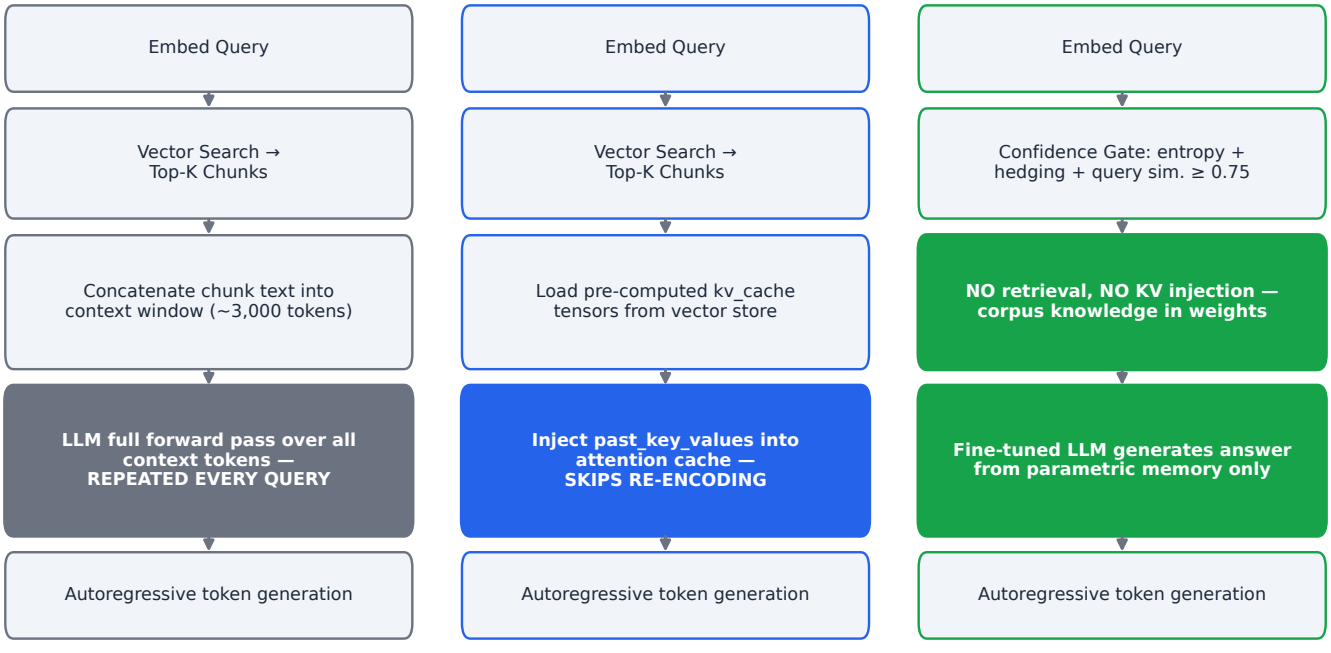
Phase 1 — Standard RAG
Embed Query
Vector Search → Top-K Chunks
Concatenate chunk text into context window (~3,000 tokens)
LLM full forward pass over all context tokens — REPEATED EVERY QUERY
Autoregressive token generation
Phase 2 — KV Injection (KVForge)
Embed Query
Vector Search → Top-K Chunks
Load pre-computed kv_cache tensors from vector store
Inject past_key_values into attention cache — SKIPS RE-ENCODING
Autoregressive token generation
Phase 3 — Parametric (KVForge)
Embed Query
Confidence Gate: entropy + hedging + query sim. ≥ 0.75
NO retrieval, NO KV injection — corpus knowledge in weights
Fine-tuned LLM generates answer from parametric memory only
Autoregressive token generation
Low confidence → fallback to Phase 2